Compliance sheet — star-lambert

Everything a supplier qualification, a legal review or an export check usually asks about star-lambert 0.1.3, in one document.

1. Identity

Document	STAR-PKG-STAR-LAMBERT-EN · v1.3 · 2026-10-16
Package	star-lambert
Version	0.1.3
Capability	star.orbit.lambert
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_lambert

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_lambert-
0.1.3-py3-none-any.whl
```

FILE	SHA-256
star_lambert-0.1.3-py3-none-any.whl	68adb79d04bdbcf1b6f4e3fdb9f455a754e6def2dd82c2df7d926d13b69ed805
star_lambert.py	4ba3d54d17e13b3267d6b010e064ef286298fa181ba23af5111c3531b93fac76
star_lambert-0.1.3-source.zip	aa47c8fb7138c39664a4d80b9e83ac6e029d5a09a7222b91a7eca42bef3581e9

5. What it does, and its limits

Zero-revolution solver for Lambert's problem (universal variables, Stumpff functions, Newton on z; Bate-Mueller-White / Curtis Algorithm 5.2). Standard library only.

- `lambert(r1, r2, tof, mu=398600.4418, prograde=True) → (v1, v2)`: velocities at r1 and r2 for a transfer of duration `tof`. Units: km, s, km/s, km^3/s^2.
- Never return an unconverged result: `RuntimeError` instead.
- Invalid input fails explicitly: `ValueError` for tof <= 0, zero position vectors, degenerate geometry (0 or 180 deg).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_lambert_0_1_3,
  author  = {Cavazzini, Andrea},
```

```
  title   = {{star-lambert}: Universal-variable Lambert orbital boundary value problem
solver},
  version = {0.1.3},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_lambert/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts